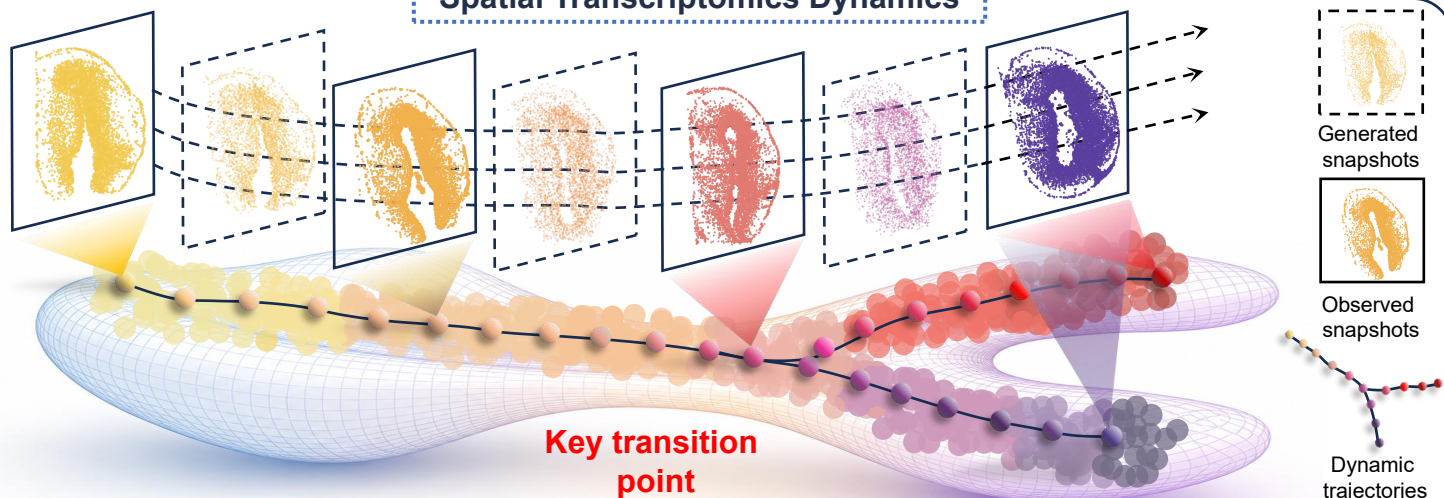
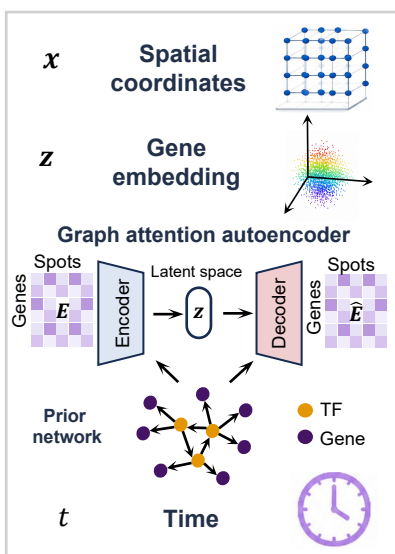
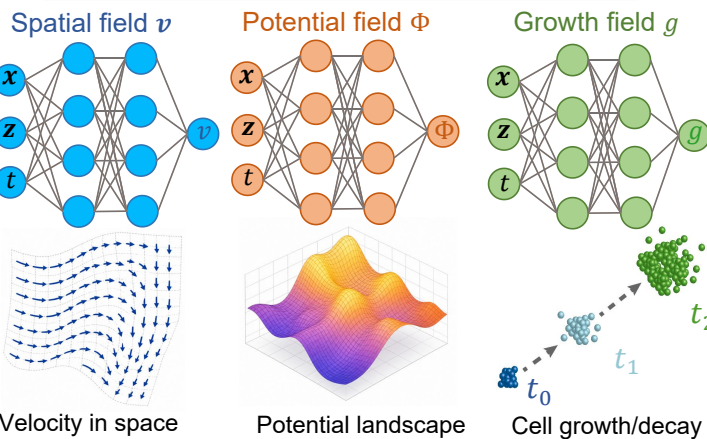
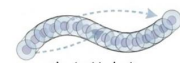
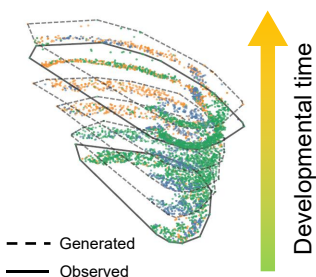
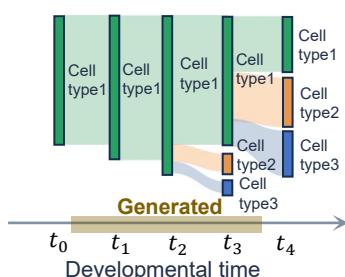
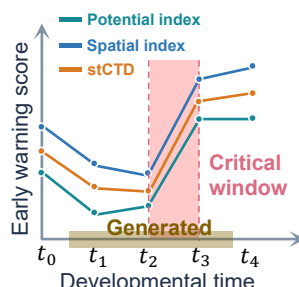


**a****Spatial Transcriptomics Dynamics****b****Training stCTD Model****Inputs****Spatiotemporal potential-transport equation**

$$\partial_t \rho + \nabla_x \cdot (\rho \mathbf{v}) - \nabla_z \cdot (\rho \nabla_z \Phi) = g \rho$$

**Optimization objective**

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{mat}} + \lambda_2 \mathcal{L}_{\text{Spa}} + \lambda_3 \mathcal{L}_{\text{phy}} + \lambda_4 \mathcal{L}_{\text{dyn}}$$

 $\mathcal{L}_{\text{mat}}$ : matching loss $\mathcal{L}_{\text{Spa}}$ : spatial loss $\mathcal{L}_{\text{phy}}$ : physical loss $\mathcal{L}_{\text{dyn}}$ : dynamic loss**c****Downstream Analyses and Applications****1 Spatial slice generation****2 Lineage inference****3 Critical point warning****4 Perturbed gene program**